

Loyalty Signalling in Hong Kong: A Textual Analysis of Political Discourse from 1997 to 2026

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Abstract

This paper investigates the evolution of “loyalty signaling” by Hong Kong Chief Executives (CEs) toward the central government. While the definition of “loyalty” remains conceptually fluid and vague in Chinese politics, we propose a six-dimensional framework to categorize how political alignment is communicated. And the authors quantify the frequency of loyalty signaling by CE’s through two proxies: (i) loyalty keywords density in speeches and press releases, and (ii) the semantic similarity between the speeches of CE’s and central leaders. By analyzing the temporal trend in loyalty signaling from 1997 to 2026, this paper reveals a distinct cyclical pattern: loyalty signaling intensifies following periods of socio-political friction or “trust crises” between Hong Kong and the Central Government. The results suggest that loyalty signaling is a reparative strategy directed to regain political confidence and signal political alignment during times of uncertainty. Notably, despite a surge in loyalty-signaling keyword density over the past decade, empirical evidence does not support the hypothesis of a growing convergence in political discourse between Hong Kong and Mainland China.

Keywords: loyalty signaling, political discourse, central-local relations, text mining

1. INTRODUCTION

Loyalty signaling is blablablablablablabla blabl blablablabla
ablablablablablablablabla

insert words here

1.1. Literature Review

insert words here

2. METHODOLOGY

2.1. Data Collection

The primary dataset consists of a corpus containing Traditional Chinese government official press releases (6994 in total) and CE speeches and articles (2697 in total). The fetched data spans five administrative terms (July 1997 to March 2026), capturing official discourse from Tung Chee-hwa, Donald Tsang, Leung Chun-ying, Carrie Lam, and John Lee.¹

The central leaders’ addresses about Hong Kong are collected as the proxy for Mainland political discourse.² This paper studies the similarity between central leader addresses and CE speeches to assess the degree of rhetorical alignment and convergence of political discourse between Mainland and Hong Kong.

2.2. Conceptualizing ‘Loyalty Signaling’

insert words here

2.3. Keyword Detection

The data cleaning phase involved adjusting character encoding for the accurate rendering of Traditional Chinese characters, handling missing values, and stripping HTML tags, punctuation,

and extraneous whitespace. A length threshold was enforced, excluding any documents with fewer than 2,000 characters. Additionally, temporal metadata was standardized by extracting publication years via regular expressions and formatting dates to DD-MM-YYYY, restricting the analytical window strictly to the 1997–2026 period.

Data wrangling procedures include transforming the unstructured corpus into structured DataFrames for quantitative analysis, and conducting text normalization and stop word removal to prepare the text for subsequent processing.

The following computational steps are performed to transform raw text into measurable density metrics:

- Temporal categorization: a regular expression function, `extract\_year`, was used to identify the year of publication from the file names.
- Keyword extraction: for every document, the script iterated through the dictionary and counted the total occurrences (Hits) of every defined keyword.
- Normalization and density metrics: to account for varying document lengths, the *Loyalty Signaling Keyword Density* was calculated using the following formula:

$$\text{Density} = \left(\frac{\sum \text{Keyword Hits}}{\text{Total Character Count}} \right) 10,000 \quad (1)$$

This normalization yields a standardized metric representing keyword frequency per 10,000 characters, allowing for a statistically valid longitudinal comparison across the dataset.

2.4. Cosine Similarity Analysis

CE speeches and Hong Kong-related discourse from Mainland Chinese leaders are parsed using regular expressions to extract publication dates and categorized into annual corpora. The raw text undergoes rigorous cleaning, including the removal of non-Chinese characters and HTML tags. The `jieba` library is used for precise Chinese semantic segmentation.

To isolate “loyalty signaling” from routine administrative language, loyalty keyword boosting and customized stop-word fil-

¹Press release for current administration (Traditional Chinese) can be found at: <https://www.ceo.gov.hk/tc/press.html>; CE speeches: <https://www.ceo.gov.hk/tc/speeches.html>; Archive of past CE’s speeches and press releases: <https://www.ceo.gov.hk/tc/links.html>

²Data regarding central leaders’ speeches about Hong Kong was accessed from the Liaison Office of the Central People’s Government (LOCPG) website: <http://www.locpg.gov.cn/zxxz/zysy.htm>. But the dataset is restrained to the period from 2008 to 2026.

ter are implemented. The TF-IDF algorithm is implemented to transform unstructured text into high-dimensional numerical vectors. The model utilizes a `max_df` of 0.85 and a `min_df` of 2 to remove extremely common or statistically insignificant tokens. The degree of discursive convergence is quantified using Cosine Similarity.

$$\text{Cosine Similarity} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

(2)

where **A** and **B** represent the TF-IDF vectors of CE speeches and central leader addresses, respectively. This metric ranges from 0 (completely dissimilar) to 1 (identical), providing a quantitative measure of rhetorical alignment over time.

3. RESULTS AND FINDINGS

3.1. Loyalty Signaling in CE Speeches

The density of loyalty-signaling keywords in Chief Executives’ speeches varied significantly across administrations (ANOVA result: $F(4,2692) = 558.02, p < .001$). Tung established a moderate baseline for loyalty rhetoric, which remained relatively stable throughout his tenure, marked by a slight uptick preceding his resignation. Following Tung, loyalty signaling plummeted. For both Tsang and Leung, the median keyword density dropped to zero, indicating that in at least half of their analyzed speeches, loyalty keywords did not appear a single time. Although Figure 1 shows a visual upward trend in signaling at the end of Leung’s term, statistical analysis demonstrates no significant difference in the overall signaling density between Tsang and Leung.

A dramatic escalation in loyalty signaling occurred during Carrie Lam’s administration. The frequency climbed steadily from 2017, reaching its peak in 2020. Her mean signaling density was more than double that of Tung’s and nearly five times higher than Leung’s. This rhetorical shift reached its absolute triumph under John Lee’s administration. Not only is his keyword density the highest, his lowest Coefficient of Variation (CV) further suggests that his high volume of loyalty signaling is not isolated to specific events; it is a standardized, highly consistent, and baked-in component of his everyday executive discourse.

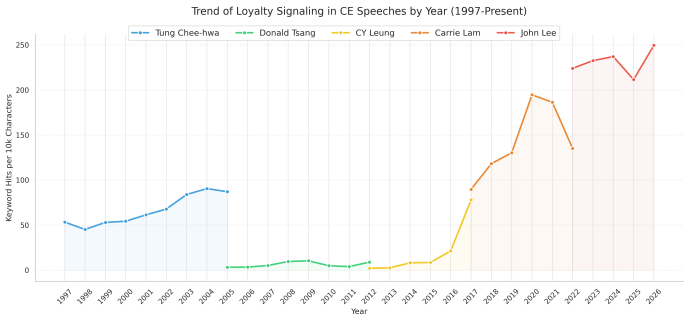


Figure 1. Density of loyalty-signaling keywords across administrations.

Table 1. Statistical Summary and Pairwise Comparison of Loyalty Keyword Density in CE Speeches

CE	n	Descriptive Statistics			Post-hoc Comparison (<i>p</i> -adj)			
		Mean	Median	CV	Tung	Tsang	Leung	Lam
Tung	497	61.15	50.98	0.84	—	—	—	—
Tsang	396	16.20	0.00	2.51	***	—	—	—
Leung	451	29.41	0.00	1.74	***	.20	—	—
Lam	646	139.93	112.40	0.76	***	***	***	—
Lee	707	228.80	211.60	0.56	***	***	***	***

Note. Overall ANOVA: $F(4,2692) = 558.02, p < .001$.
*** $p < .001$; ** $p < .01$; * $p < .05$.

Chronologically, the year of 2015 serves as a critical inflection point, representing the immediate rhetorical response to the 2014 Umbrella Movement. This shift was preceded by the State Council’s June 2014 White Paper, which systemically introduced the doctrine of “Comprehensive Jurisdiction” for the first time.³ This trajectory of intensified signaling reached its peak in 2020, following the Anti-Extradition Law Amendment Bill Movement . While the promulgation of the National Security Law (NSL) in 2020 was followed by a temporary dip in signaling density, the intensity subsequently surged to an all-time high during John Lee’s administration. These patterns strongly support the conclusion that loyalty signaling in the HKSAR is most acute during periods of peak political uncertainty and serves as a reparative mechanism to restore central-local confidence.

While the overall trend for “loyalty signaling” shows a statistically significant increase during the tenures of Carrie Lam and John Lee, this pattern is not uniform across all categories. First of all, there is no rhetorical loyalty signaling throughout all the periods. Secondly, ideological loyalty has constantly remained the most dominant category in terms of absolute volume, reaching a density of 120.4 per 10,000 characters in Lee’s administration. But it shall be noted that Tung Chee-hwa also exhibited a high propensity for ideological signaling; as the first Chief Executive, he had to frequently articulate the relationship between the HKSAR and the mainland’s socialist system. Thirdly, none of the five CEs demonstrated a significant enthusiasm for “personal loyalty” signaling, indicating that political alignment in the HKSAR is projected as an institutional or systemic commitment rather than an individual allegiance.

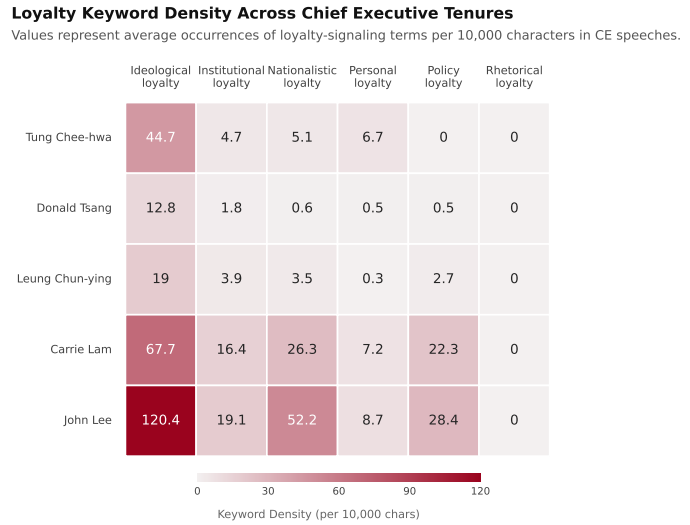


Figure 2. Density of loyalty-signaling keywords by category across administrations.

3.2. Loyalty Signaling in Press Releases

For Tung Chee-hwa, Carrie Lam, and John Lee, there is a statistically significant difference ($p < .001$) between their spoken speeches and written press releases. In all three cases, they deploy vastly more loyalty signaling through spoken words than in press releases. Whereas for Tsang and Leung, such gap is absent. The Welsh t-test results indicate no statistically significant difference between what they say and what their administrations publish. When overall loyalty signaling was low, it was uniformly

³See https://www.hmo.gov.cn/xwzx/zwyw/202312/t20231208_38591.html This document signaled Beijing’s dissatisfaction with the local administration’s perceived inability to manage the precursors to “Occupy Central,” emphasizing that Hong Kong’s autonomy is a delegated power rather than an inherent right.

low across both channels.

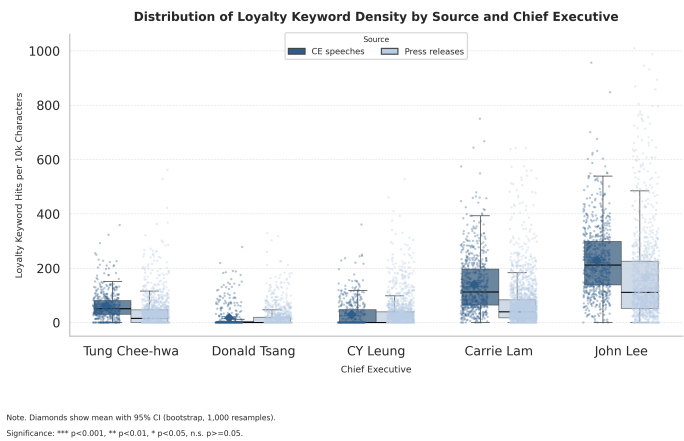


Figure 3. Comparison of loyalty-signaling keyword density between CE speeches and press releases.

Table 2. Comparison of Loyalty Keyword Density (Welch’s <i>t</i> -test)						
CE	Speech <i>n</i>	Press <i>n</i>	Mean (S)	Mean (P)	<i>t</i> -stat	<i>p</i> -value
Tung	499	1338	60.90	34.46	9.60	< .001***
Tsang	423	1105	17.43	17.39	0.02	.99
Leung	451	1603	29.41	31.47	-0.74	.46
Lam	646	1878	139.93	67.74	15.59	< .001***
Lee	707	1070	228.80	165.14	9.02	< .001***

Note. Mean (S) = Speeches; Mean (P) = Press Releases. Welch’s *t*-test was used due to unequal variances and sample sizes. *** $p < .001$; ** $p < .01$; * $p < .05$.

There is a notable shift in the Lee administration’s rhetoric turning into 2026. For the first time since 1997, the data reveal an inversion in the signaling medium, with significantly more loyalty being signaled through official press releases than through the Chief Executive’s personal speeches. While it remains unclear whether this marks an outlier or a strategic realignment in public discourse, two hypotheses offer potential explanations for this phenomenon. The first hypothesis is that, having spent 2022–2025 establishing an undeniably high baseline of political alignment, the administration may now consider that trust secured. The burden of loyalty signaling is being pushed down into the civil service machinery through the standardized, boilerplate text of daily press releases. Another hypothesis is that the administration, cautious of potential backlash from civil society and the commercial sector—where repeated loyalty signaling might shake confidence in Hong Kong’s autonomy—the administration may be intentionally shifting these signals into less visible bureaucratic channels. This allows the Chief Executive to reserve his highly visible public speeches for pragmatic, economically oriented messaging.

3.3. Textual Similarity

The consistently low overall textual similarity between Chief Executive speeches and state leaders’ Hong Kong-related addresses highlights a crucial finding: loyalty signaling has not driven a wholesale assimilation into Mainland political rhetoric. This demonstrates that the recent rise in loyalty signaling has not fundamentally changed the underlying structure of Hong Kong’s political discourse. Despite a higher frequency of specific political keywords, there is no empirical evidence to suggest that the CE has adopted the generalized “officialese” often characteristic of Mainland bureaucratic texts.

Notably, contrary to the hypothesis that John Lee might heavily utilize Mainland-style bureaucratic jargon to approximate Bei-

jing’s political discourse, it appears from the data that he employs a highly targeted, “modular” signaling strategy. Rather than structurally adopting Mainland-style rhetorical style, Lee strategically inserts specific loyalty-signaling keywords (such as “national security” and “patriots administering Hong Kong”) with high frequency and precision to establish an unambiguous political baseline. However, his overarching narrative style, daily administrative vocabulary, and linguistic structure have not converged with the Mainland’s official speech system. This represents a highly pragmatic rhetorical approach: it guarantees strict political compliance while actively preserving the localized, pragmatic, and technocratic linguistic framework inherent to Hong Kong’s executive branch.

4. DISCUSSION

5. CONCLUSION

6. BIBLIOGRAPHY

REFERENCES

[1] Christian Feuersänger. (2024). PGFPlots: A LaTeX package to create normal/logarithmic plots in two and three dimensions. <https://ctan.org/pkg/pgfplots>

7. APPENDIX

8. FIGURES

spanishFig.Fig. ?? shows a 3D surface visualization of a bivariate function.

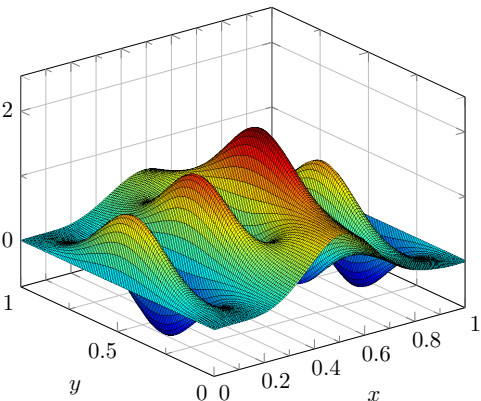


Figure 4. 3D surface obtained from PGFPlots [PGFPlots].

9. TABLES

spanishTablaTable ?? shows an example table of some astronomical objects.

Table 3. Astronomical Object Data		
Object	Type	Distance (Light Years)
Alpha Centauri	Star system	4.37
Betelgeuse	Red supergiant star	642.5
Andromeda	Spiral galaxy	2.537 million
Earth	Planet	0
Sirius	Binary star system	8.6

Note: The table contains data of some famous celestial objects.

The `\tabletext {}` command is used to add notes to tables easily.

10. DOCUMENT DATA

10.1. Front-Matter Footer Block*

A new block called `\docinfo {}`, has been added to the bottom right of the first page, acting as a floating element which stays fixed in place. This replaces the corresponding author block previously located above the abstract in earlier versions.

The `\docinfo {}` block was created to give you greater control over supplementary document data, such as publication dates, licensing information, extended author details, and other front-matter notes. You can fully customize its content to suit your needs.

Additionally, the new `\corres {}` command serves as a flexible alternative to `\thanks {}`. Since modifying the standard `\thanks {}` command can be complex and its positioning is often rigid, `\corres {}` was introduced to offer a more adaptable solution without sacrificing functionality.

spanishNotaNote

You can use `\corres{}` as many times as needed since it automatically ensures that each entry remains correctly associated. For best results, I recommend using text symbols rather than mathematical ones (e.g., `\textasteriskcentered`, `\textdagger`, etc.).

If `\corres {}` is not used, `\docinfo {}` will automatically adjust its position. The implementation was designed to handle any combination of front-matter elements you might need when starting your document. In this example template, you'll find a minimal demonstration that combines both to illustrate their interaction.

10.2. Headers and Footers*

10.2.1. Headers

On all pages except the first, the document title and journal appear in the header. Since twoside layout is enabled, its position alternates depending on whether the page is odd or even.

10.2.2. Footers

In the first page, six elements display supplementary information for your document:

- `\journalname {}` — Displays the journal name, university, institute, company, or any institutional affiliation.
- `\theday {}` — Displays the current date.
- `\vol {}` — Specifies the volume number of the publication.
- `\no {}` — Specifies the issue number within the volume.
- `\journal {}` — Defines the full journal title.
- `\thepage` — Shows the current page number and the total count.

In all cases, you can freely rearrange or customize the order of these elements by modifying the class file to suit your needs. If any of them are not required, simply omit the corresponding command — the remaining elements will automatically adjust their spacing to remain evenly distributed.

11. RHO CUSTOM PACKAGES

11.1. Rhobabel.sty

This package have all the commands that automatically translate from English to Spanish when this custom package is defined.

By default, this document has its content in English. However, at the beginning of the document you will find a recommendation when writing in Spanish.

You may modify this package if you want to use other language than English or Spanish. This will make easier to translate your document without having to modify the class document.

11.2. Rhoenvs.sty

This package provides custom environments designed to enhance the visual presentation and structure of your document. Key examples include `rhoenv`, `info`, and `note`.

Their style is defined by `rhoenvstyle` — allowing you to customize their appearance according to your document's requirements.

An example using the `rhoenv` environment is shown below.

Custom Title

This is an example of the custom title environment. To add a title type this command `[frametitle=Custom Title]` next to the beginning of this environment (as shown in this example).

The `info` and `note` are environments which have a predefined title and translate their title to Spanish automatically when this language is defined.

12. EQUATIONS

`spanishEcuaciónEquation ??` shows the Schrödinger equation as an example.

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \left(\frac{\hbar^2}{2m} (\nabla^2) + V(\mathbf{r}, t) \right) \Psi(\mathbf{r}, t) \quad (3)$$

The `amssymb` and `amsmath` packages were not required, as STIX2 font incorporates mathematical symbols for writing quality equations.

spanishNotaNote

If you would like to change the values that adjust the spacing above and below the equations, change the `\eqskip` value until the preferred spacing is set. The default value is set to 9pt.

13. CODES

13.1. Coding with Minted*

The `minted` package offers customized features for adding codes. In addition, the template is designed to work seamlessly with Fira Code, a monospaced font specially crafted for programming environments.

If your are using a desktop app as TeXstudio, try these steps to make this package work on Windows:

1. Install Python — A stable version (e.g., v.3.11)
2. Open the terminal and type `pip install Pygments`.
3. Update if a newer version is available.
4. Go to TeXStudio settings and change the default compiler to `pdflatex -shell-escape -interaction=nonstopmode \%.tex`.
5. Compile and wait for the result.

Caution

Ensure that `pygments` is properly installed and added to your system's PATH. Otherwise, you may encounter compilation errors. Additionally, enable shell escape when compiling, as it is required for `minted` to process and highlight code.

In Overleaf, this package is easier to use since it does not require any pdflatex installations or modifications — just add the code

as shown in the example. `spanishCódigoCode ??` shows a Python example.

```
def greet(name: str) -> None:
    print(f"Hello, {name}.")

def main() -> None:
    greet("world")

if __name__ == "__main__":
    main()
```

spanishCódigoCode 1. Python code example with listings.

You can customize its design changing `\usemintedstyle` command. The different styles offered by the `minted` package can be preview through this link — <https://pygments.org/styles/>.

13.2. Coding with Listings

Since `minted` requires additional installations and can be complex in some desktop $\text{\TeX}$ editors, you can use the `listings` package instead, which provides a simpler way to include code.

```
def greet(name: str) -> None:
    print(f"Hello, {name}.")

def main() -> None:
    greet("world")

if __name__ == "__main__":
    main()
```

spanishCódigoCode 2. Python code example with listings.

While `listings` is a simpler and widely supported package for code formatting, `minted` offers a more modern and powerful approach. It leverages the Pygments syntax highlighter to deliver superior coloring, language support, and styling options. One of its key advantages is the ability to easily switch between built-in color styles.

In contrast, `listings` requires manual setup for colors, fonts, and formatting rules. For users who prefer fine-tuned customization, the styling options for `listings` are organized in `rho.cls` file and can be modified at any time to suit individual preferences.

13.3. Inline Code*

As shown in this template, inline code has a custom style for improved visual appeal. To use it, simply add `\inlinecode {}` command and place your code inside the braces.

■ NUMBERLESS SECTION

Declaring a numberless section automatically inserts a square symbol before the title. This unique feature is specific to first-level sections within this class.

As this symbol also affects the table of contents and references titles, users who prefer to disable it may edit the section styling parameters in `rho.cls`.

14. BIBLIOGRAPHY

Bibliography management is handled by `biblatex`, with the default citation and reference style set to IEEE. The `citestyle=numeric-comp` option is enabled, allowing multiple citations to be grouped within a single bracket (e.g.,

[`davis2023signalflow`, `bell2022codefont`]). The citation format can be customized directly in `rho.cls` to suit any preferred style.

■ FAQ

How do I manage my references?

To manage your references, I recommend using the tool `scribbr`. You can simply enter the URL or create your own citation, and then export it to $\text{\LaTeX}$ using the options in the three-dot menu. The generated citation can be copied and pasted into `rho.bib`, the file designated for bibliography management. You may rename this file, but if you do, remember to update the `\adddbibresource` command in `rho.cls` under the `biblatex` section.

spanishNotaNote

Some platforms, such as Google Scholar or scientific journals, provide citations directly in BibTeX format. Therefore, check if there is a “how to cite this document” section to streamline the citation process even further.

If you have any further questions, you can refer to the following page — [Bibliography management with biblatex](#).

What should I do with the example files?

The template includes sample content — such as an example figure, bibliography entries, and the `example.py` script — to demonstrate how the layout and features work. Once you start customizing the document for your own use, feel free to delete all example files and entries you don’t need.

How do I place equations easily?

For equations, we have two options: inline or on its own line. For inline equations, simply place a dollar sign (\$) at the beginning and end of the equation. However, if you want the equation to be displayed on its own line, you need to use the equation environment.

If you find it challenging to write formulas directly in $\text{\LaTeX}$, you can use text editors like Word. In the equations menu, you can select $\text{\LaTeX}$ in the conversion section and copy and paste the equation you wrote into one of these environments.

How do I change the paper size?

By default, this class was adapted for `a4paper` and test it with `letterpaper`. The following paper sizes are available in $\text{\LaTeX}$:

- `letterpaper` (11 8.5 in)
- `legalpaper` (14 8.5 in)
- `executivepaper` (10.5 7.25 in)
- `a4paper` (21 29.7 cm)
- `a5paper` (21 14.8 cm)
- `b5paper` (25 17.6 cm)

■ CONTACT ME

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■ GITHUB REPOSITORY

Visit the repository to access the source code, track ongoing development, report issues, and stay up to date with the latest changes.

 <https://github.com/MemoJimenez/Rho-class>